



Time & Attendance Management System

Complete Project Plan & Wireframe Specifications

Project Type: Full-Stack Web Application

Stack: Fullstack JavaScript (React + Express + PostgreSQL)

Primary Users: Employees, Managers, HR Administrators

Version: 1.0 - Initial Planning Phase

Date: 3/2/2026



Executive Summary

A comprehensive time and attendance management system that enables employees to clock in/out, request time off, and view their attendance records. Managers can approve requests, run reports, and monitor team attendance. HR administrators have full system oversight with advanced reporting and configuration capabilities.



Core Objectives

- **Simplify Time Tracking:** Provide an intuitive interface for employees to record work hours
- **Streamline Approvals:** Enable managers to efficiently review and approve time-off requests
- **Enhance Visibility:** Give all stakeholders real-time access to attendance data
- **Ensure Compliance:** Maintain accurate records for payroll and regulatory requirements
- **Improve Productivity:** Reduce administrative overhead through automation



User Roles & Permissions

1. Employee

- Clock in/out for shifts
- View personal attendance history
- Submit time-off requests (vacation, sick leave, personal days)
- View request status and approval history
- Access personal time balance (PTO, sick days)

2. Manager

- All employee capabilities
- View team attendance in real-time
- Approve/deny time-off requests from team members
- Run team attendance reports
- View team time-off calendar
- Receive notifications for pending approvals

3. HR Administrator

- All manager capabilities across all departments
- Manage user accounts and role assignments
- Configure time-off policies and accrual rules
- Access company-wide attendance reports

- Export data for payroll processing
- System configuration and settings management

MVP Features (Phase 1)

1. User Authentication with Replit Auth

- Secure login/logout functionality
- Role-based access control (Employee, Manager, HR Admin)
- Session management

2. Clock In/Out System

- One-click clock in/out button with timestamp
- Visual indicator of current status (clocked in/out)
- Automatic calculation of hours worked
- Today's shift summary display

3. Attendance History

- Personal attendance record table with date, clock in/out times, total hours
- Filter by date range
- Sort by various columns
- Weekly and monthly hour summaries

4. Time-Off Request Management

- Request form with date range picker, type selection, and notes
- Visual calendar showing requested and approved time off
- Status tracking (pending, approved, denied)
- Request history table

5. Manager Approval Dashboard

- Pending requests queue with employee details
- Quick approve/deny actions with optional comments
- Team attendance overview
- Notification badges for pending items

6. Basic Reporting

- Employee attendance report (personal)
- Team attendance report (for managers)
- Company-wide report (for HR)
- Export to CSV functionality

7. Responsive Dashboard Design

- Clean, professional interface with Shadcn UI components
- Mobile-responsive layout
- Intuitive navigation with role-based menu items
- Real-time status updates



Next Phase Features (Phase 2)

1. Manual Time Entry Corrections

- Employees request corrections for missed punches
- Manager approval workflow for corrections

2. Advanced Time-Off Policies

- Automatic PTO accrual calculations
- Multiple leave types with separate balances
- Blackout date configuration

3. Email Notifications

- Request submission confirmations
- Approval/denial notifications
- Reminder emails for pending manager actions

4. Enhanced Reporting & Analytics

- Overtime tracking and alerts
- Attendance trends and patterns
- Custom report builder with filters
- Scheduled report generation

5. Department & Team Management

- Hierarchical organization structure
- Department-specific policies
- Team calendar views



Technology Stack

Frontend

- React 18 (SPA)
- TypeScript
- Tailwind CSS
- Shadcn UI
- React Query
- Wouter (routing)
- React Hook Form
- date-fns

Backend

- Node.js + Express
- TypeScript
- Replit Auth
- PostgreSQL
- Drizzle ORM
- Express Session

Database Tables

- users (Replit Auth)
- user_roles
- attendance_records
- time_off_requests
- time_off_balances

Dev Tools

- Vite
- Drizzle Kit
- ESLint
- Prettier
- Replit Deployment

Wireframes

1. Login Page

Time & Attendance System

 Login with Replit

Secure authentication powered by Replit Auth

✓ Track your time • ✓ Manage requests • ✓ Stay organized

2. Employee Dashboard

Navigation

-  Dashboard
-  My Attendance
-  Time Off
-  Profile

John Doe
Employee

Dashboard

Current Status: ● Clocked Out

CLOCK IN

Today's Hours

0.0
hrs

This Week

32.5
hrs

PTO Balance

-  Vacation: 10 days
-  Sick Leave: 5 days

Recent Activity

Yesterday	8:00 AM - 5:00 PM	8.0 hrs
2 days ago	8:15 AM - 5:15 PM	8.0 hrs

3. My Attendance History

Attendance History

 Date Range: Last 30 Days ▼

Summary

Total Hours	Days Worked	Daily Average
160.0	20	8.0

Date	Clock In	Clock Out	Break	Total Hours	Status
03/01/2026	8:00 AM	5:00 PM	1:00 hr	8.0 hrs	✓ Complete
02/28/2026	8:15 AM	5:15 PM	1:00 hr	8.0 hrs	✓ Complete
02/27/2026	8:00 AM	6:00 PM	1:00 hr	9.0 hrs	⚠ Overtime

 Export to CSV

4. Time Off Request

New Request

Request Type

[Dropdown: 🏖️ Vacation / 🤒 Sick / 📄 Personal]

Start Date

[📅 Date Picker]

End Date

[📅 Date Picker]

Notes (Optional)

[Text Area - multiline input]

📅 Days Requested: 3
Remaining Balance: 7 days

Submit Request

My Requests

PENDING

Mar 15-17, 2026
🏖️ Vacation • 3 days
⌚ Awaiting manager approval

APPROVED

Feb 10-12, 2026
🏖️ Vacation • 3 days
✓ Approved by Jane Smith

APPROVED

Jan 5, 2026
🤒 Sick Leave • 1 day
✓ Approved by Jane Smith

Team Calendar

[Visual Calendar Grid View]

- Legend:
- Your scheduled time off
 - Team members' time off

5. Manager Dashboard

Navigation

-  Dashboard
-  My Attendance
-  My Time Off
-  Team View
-  Approvals 3
-  Reports

Jane Smith

Manager

Team Overview

 Pending Approvals: 3

Action required

Team Size

12

Clocked In

8

On Leave

2

Team Status			
Employee	Status	Today	This Week
John Doe	Clocked Out	0.0 hrs	32.5 hrs
Alice Johnson	Clocked In (8:00 AM)	3.5 hrs	28.0 hrs
Bob Williams	On Leave	-	24.0 hrs

6. Manager Approval Queue

Pending Approvals (3)

John Doe - Vacation Request

March 15-17, 2026 (3 days)

Submitted: March 1, 2026

"Family trip to the beach"

✓ Approve

✗ Deny

Alice Johnson - Sick Leave

March 2, 2026 (1 day)

Submitted: March 2, 2026

"Not feeling well"

✓ Approve

✗ Deny

Recently Processed

Employee	Type	Dates	Status	Processed By	Date
Sarah Lee	Vacation	Feb 20-22	✓ Approved	You	Feb 15
Mike Chen	Sick Leave	Feb 10	✓ Approved	You	Feb 10

7. HR Administrator Dashboard

Navigation

-  Dashboard
-  My Attendance
-  My Time Off
-  Company
-  Users
-  Settings
-  Reports

Admin User

HR Admin

Company Overview

Total Employees	Active Now	On Leave	Pending
127	94	8	15

Department Breakdown

Department	Employees	Active	On Leave	Avg Hrs/Wk
Engineering	45	38	3	40.2
Sales	32	28	2	38.5
Marketing	18	15	1	39.8

Quick Actions

-  Generate Payroll Report
-  Export All Data
-  Add New Employee

Recent Activity

- 3 new time-off requests
- 5 requests approved today
- 2 new employees added

8. Reports & Analytics

Generate Report

Report Type

[Dropdown: Employee / Team / Company]

Date Range

[📅 From - To Date Picker]

Export Format

[Dropdown: CSV / PDF / Excel]

Optional Filters

Department: [All ▼]

Employee: [Search... 🔍]

Status: [All ▼]

 Generate Report

Report Preview

Employee	Department	Total Hours	Days Worked	Days Off	Overtime
John Doe	Engineering	160.0	20	2	4.5
Alice Johnson	Sales	152.0	19	3	0.0
Bob Williams	Marketing	168.0	21	1	8.0



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Email Report



Database Schema

users

- id (PK, UUID)
- email (unique)
- firstName
- lastName
- profileImageUrl
- createdAt
- updatedAt

attendance_records

- id (PK)
- userId (FK)
- clockIn (timestamp)
- clockOut (timestamp)
- breakDuration
- totalHours
- date
- notes

user_roles

- id (PK)
- userId (FK)
- role (enum)
- departmentId
- createdAt

time_off_requests

- id (PK)
- userId (FK)
- requestType
- startDate
- endDate
- totalDays
- status
- reviewedBy (FK)
- reviewedAt
- notes

time_off_balances

- id (PK)
- userId (FK)
- year
- vacationDays
- sickDays
- personalDays



Security & Best Practices

- **Authentication:** Replit Auth (OAuth-based)
- **Authorization:** Role-based access control on all endpoints
- **Data Privacy:** Users only see authorized data
- **Session Security:** Secure cookies, HTTPS only
- **Input Validation:** Client and server-side validation
- **SQL Injection Prevention:** Parameterized queries via Drizzle ORM



Estimated Timeline

- Week 1-2:** Setup, auth, database, UI framework
- Week 3-4:** Clock in/out, attendance history
- Week 5-6:** Time-off requests, approval workflow
- Week 7:** Manager dashboard, team views
- Week 8:** HR features, reporting
- Week 9:** Testing, bug fixes, polish
- Week 10:** Deployment, training, docs

Time & Attendance Management System

Project Plan & Wireframe Document v1.0

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This document contains the complete project scope, technical specifications, and visual wireframes.